

UNIT-II

VECTOR CALCULUS

Directional derivative

The derivative of a point function (scalar or vector) in a particular direction is called its directional derivative along the direction.

The directional derivative of a scalar point function ϕ in a given direction is the rate of change of ϕ in the direction. It is given by the component of $\text{grad}\phi$ in that direction.

The directional derivative of a scalar point function

$\phi(x, y, z)$ in the direction of $\vec{a}$ is given by $\frac{\nabla\phi \cdot \vec{a}}{|\vec{a}|}$.

Directional derivative of ϕ is maximum in the direction of $\nabla\phi$.

Hence the maximum directional derivative is $|\nabla\phi|$ or $|\text{grad}\phi|$

Unit normal vector to the surface

If $\phi(x, y, z)$ be a scalar function, then $\phi(x, y, z) = c$ represents a surface and the unit normal vector to the surface ϕ is given by

$$\frac{\nabla\phi}{|\nabla\phi|}$$

Equation of the tangent plane and normal to the surface

Suppose $\vec{a}$ is the position vector of the point (x_0, y_0, z_0)

On the surface $\phi(x, y, z) = c$. If $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$ is the position vector of any point (x, y, z) on the tangent plane to the surface at $\vec{a}$, then the equation of the tangent plane to the surface ϕ at a given point $\vec{a}$ on it is given by $(\vec{r} - \vec{a}) \cdot \text{grad}\phi = 0$

If $\vec{r}$ is the position vector of any point on the normal to the surface at the point $\vec{a}$ on it. The vector equation of the normal at a given point $\vec{a}$ on the surface ϕ is $(\vec{r} - \vec{a}) \times \text{grad}\phi = 0$

The Cartesian form of the normal at (x_0, y_0, z_0) on the surface

$\phi(x, y, z) = c$ is

$$\frac{x - x_0}{\frac{\partial\phi}{\partial x}} = \frac{y - y_0}{\frac{\partial\phi}{\partial y}} = \frac{z - z_0}{\frac{\partial\phi}{\partial z}}$$

Divergence of a vector

If $\vec{F}(x, y, z)$ is a continuously differentiable vector point function in a given region of space, then the divergences of $\vec{F}$ is defined by

$$\nabla \cdot \vec{F} = \text{div } \vec{F} = \vec{i} \frac{\partial F_x}{\partial x} + \vec{j} \frac{\partial F_y}{\partial y} + \vec{k} \frac{\partial F_z}{\partial z}$$

$$= \sum \vec{i} \frac{\partial \vec{F}}{\partial x}$$

If $\vec{F} = F_1 \vec{i} + F_2 \vec{j} + F_3 \vec{k}$, then $\text{div } \vec{F} = \nabla \cdot (F_1 \vec{i} + F_2 \vec{j} + F_3 \vec{k})$

$$\text{i.e., } \text{div } \vec{F} = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}$$

Solenoidal Vector

A vector $\vec{F}$ is said to be solenoidal if $\text{div } \vec{F} = 0$ (ie) $\nabla \cdot \vec{F} = 0$

Curl of vector function

If $\vec{F}(x, y, z)$ is a differentiable vector point function defined at each point (x, y, z) , then the curl of $\vec{F}$ is defined by

$$\begin{aligned} \text{curl } \vec{F} &= \nabla \times \vec{F} \\ &= \vec{i} \times \frac{\partial \vec{F}}{\partial x} + \vec{j} \times \frac{\partial \vec{F}}{\partial y} + \vec{k} \times \frac{\partial \vec{F}}{\partial z} \\ &= \sum \vec{i} \times \frac{\partial \vec{F}}{\partial x} \end{aligned}$$

If $\vec{F} = F_1 \vec{i} + F_2 \vec{j} + F_3 \vec{k}$, then $\text{curl } \vec{F} = \nabla \times (F_1 \vec{i} + F_2 \vec{j} + F_3 \vec{k})$

$$\begin{aligned} \text{curl } \vec{F} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{vmatrix} \\ &= \vec{i} \left(\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z} \right) - \vec{j} \left(\frac{\partial F_3}{\partial x} - \frac{\partial F_1}{\partial z} \right) + \vec{k} \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right) \end{aligned}$$

Curl $\vec{F}$ is also said to be rotation $\vec{F}$

Irrotational Vector

A vector $\vec{F}$ is called irrotational if $\text{Curl } \vec{F} = 0$

(ie) if $\nabla \times \vec{F} = 0$

Scalar Potential

If $\vec{F}$ is an irrotational vector, then there exists a scalar function ϕ

Such that $\vec{F} = \nabla \phi$. Such a scalar function is called scalar potential of $\vec{F}$

Properties of Gradient

1. If f and g are two scalar point function that $\nabla(f \pm g) = \nabla f \pm \nabla g$ (or)

$$\text{grad } (f \pm g) = \text{grad } f \pm \text{grad } g$$

$$\text{Solution: } \nabla(f \pm g) = \left(\vec{i} \frac{\partial}{\partial x} + \vec{j} \frac{\partial}{\partial y} + \vec{k} \frac{\partial}{\partial z} \right) (f \pm g)$$

$$\begin{aligned}
&= \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) (f \pm g) \\
&= i \frac{\partial f}{\partial x} \pm i \frac{\partial g}{\partial x} + j \frac{\partial f}{\partial y} \pm j \frac{\partial g}{\partial y} + k \frac{\partial f}{\partial z} \pm k \frac{\partial g}{\partial z} \\
&= \left(i \frac{\partial f}{\partial x} + j \frac{\partial f}{\partial y} + k \frac{\partial f}{\partial z} \right) \pm \left(i \frac{\partial g}{\partial x} + j \frac{\partial g}{\partial y} + k \frac{\partial g}{\partial z} \right) \\
&= \nabla f \pm \nabla g
\end{aligned}$$

- 2 If f and g are two scalar point functions then $\nabla(fg) = f\nabla g + g\nabla f$ (or)
 $\text{grad}(fg) = f\text{grad}g + g\text{grad}f$

Solution: $\nabla(fg) = \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) (fg)$

$$\begin{aligned}
&= i \frac{\partial}{\partial x} (fg) + j \frac{\partial}{\partial y} (fg) + k \frac{\partial}{\partial z} (fg) \\
&= i \left(f \frac{\partial g}{\partial x} + g \frac{\partial f}{\partial x} \right) + j \left(f \frac{\partial g}{\partial y} + g \frac{\partial f}{\partial y} \right) + k \left(f \frac{\partial g}{\partial z} + g \frac{\partial f}{\partial z} \right) \\
&= f \left(i \frac{\partial g}{\partial x} + j \frac{\partial g}{\partial y} + k \frac{\partial g}{\partial z} \right) + g \left(i \frac{\partial f}{\partial x} + j \frac{\partial f}{\partial y} + k \frac{\partial f}{\partial z} \right) \\
&= f\nabla g + g\nabla f
\end{aligned}$$

- 3 If f and g are two scalar point function then $\nabla \cdot \frac{f}{g} = \frac{g\nabla f - f\nabla g}{g^2}$ where
 $g \neq 0$

Solution: $\nabla \cdot \frac{f}{g} = \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) \cdot \frac{f}{g}$

$$\begin{aligned}
&= \sum i \frac{\partial}{\partial x} \left(\frac{f}{g} \right) \\
&= \sum i \frac{g \frac{\partial f}{\partial x} - f \frac{\partial g}{\partial x}}{g^2} \\
&= \frac{1}{g^2} g \sum i \frac{\partial f}{\partial x} - f \sum i \frac{\partial g}{\partial x} \\
&= \frac{1}{g} [g\nabla f - f\nabla g]
\end{aligned}$$

- 4 If $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$ such that $r = |\vec{r}|$, prove that $\nabla r^n = nr^{n-2}\vec{r}$

Solution: $\nabla r^n = \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) r^n$

$$= i \frac{\partial r^n}{\partial x} + j \frac{\partial r^n}{\partial y} + k \frac{\partial r^n}{\partial z}$$

$$\begin{aligned}
&= i nr \frac{\partial r}{\partial x} + j nr \frac{\partial r}{\partial y} + k nr \frac{\partial r}{\partial z} \\
&= nr^{n-1} \left[x \frac{\partial r}{\partial x} + y \frac{\partial r}{\partial y} + z \frac{\partial r}{\partial z} \right] \\
&= \frac{nr^{n-1}}{r} (x i + y j + z k) \\
&= \frac{nr^{n-1}}{r} r
\end{aligned}$$

5 Find a unit normal to the surface $x^2y + 2xz = 4$ at $(2, -2, 3)$

Solution: Given that $\phi = x^2y + 2xz$

$$\begin{aligned}
\nabla\phi &= \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) (x^2y + 2xz) \\
&= (2xy + 2z)i + (x^2)j + (2x)k
\end{aligned}$$

At $(2, -2, 3)$

$$\begin{aligned}
\nabla\phi &= (8 + 6)i + 4j + 4k \\
&= -2i + 4j + 4k
\end{aligned}$$

$$|\nabla\phi| = \sqrt{4 + 16 + 16} = \sqrt{36} = 6$$

Unit normal to the given surface at $(2, -2, 3)$

$$\begin{aligned}
\frac{\nabla\phi}{|\nabla\phi|} &= \frac{-2i + 4j + 4k}{6} \\
&= \frac{1}{3}(-i + 2j + 2k)
\end{aligned}$$

6 Find the directional derivative of $\phi = x^2yz + 4xz^2 + xyz$ at $(1, 2, 3)$ in the direction of $2i + j - k$

Solution: Given $\phi = x^2yz + 4xz^2 + xyz$

$$\begin{aligned}
\nabla\phi &= \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) (x^2yz + 4xz^2 + xyz) \\
&= (2xyz + 4z^2 + yz)i + (x^2z + xz)j + (x^2y + 8xz + xy)k
\end{aligned}$$

At $(1, 2, 3)$

$$\nabla\phi = 54i + 6j + 28k$$

Given: $a = 2i + j - k$

$$\therefore |a| = \sqrt{4 + 1 + 1} = \sqrt{6}$$

$$\begin{aligned}
 \therefore D.D &= \nabla \phi \cdot \frac{\vec{a}}{|\vec{a}|} \\
 &= \frac{1}{\sqrt{6}} [54i + 6j + 28k] \cdot \frac{2\vec{i} + 2\vec{j} - \vec{k}}{\sqrt{6}} \\
 &= \frac{1}{\sqrt{6}} [108 + 6 - 28] = \frac{1}{\sqrt{6}} [86]
 \end{aligned}$$

7. Find the angle between the surface $x^2 + y^2 + z^2 = 5$ and $x^2 + y^2 + z^2 - 2x = 5$ at $(0, 1, 2)$

Solution: Let $\phi_1 = x^2 + y^2 + z^2$ and $\phi_2 = x^2 + y^2 + z^2 - 2x$

$$\begin{aligned}
 \frac{\partial \phi_1}{\partial x} &= 2x, \quad \frac{\partial \phi_1}{\partial y} = 2y, \quad \frac{\partial \phi_1}{\partial z} = 2z \\
 \frac{\partial \phi_2}{\partial x} &= 2x - 2, \quad \frac{\partial \phi_2}{\partial y} = 2y, \quad \frac{\partial \phi_2}{\partial z} = 2z \\
 \nabla \phi_1 &= 2x\vec{i} + 2y\vec{j} + 2z\vec{k} \\
 \nabla \phi_2 &= (2x - 2)\vec{i} + 2y\vec{j} + 2z\vec{k}
 \end{aligned}$$

At $(0, 1, 2)$

$$\begin{aligned}
 \nabla \phi_1 &= 2\vec{j} + 4\vec{k} \\
 \nabla \phi_2 &= -2\vec{i} + 2\vec{j} + 4\vec{k} \\
 \cos \theta &= \frac{\nabla \phi_1 \cdot \nabla \phi_2}{|\nabla \phi_1| |\nabla \phi_2|} = \frac{2\vec{j} + 4\vec{k} \cdot -2\vec{i} + 2\vec{j} + 4\vec{k}}{\sqrt{4 + 16} \sqrt{4 + 4 + 16}} \\
 \cos \theta &= \frac{4 + 16}{\sqrt{20} \sqrt{24}} = \frac{20}{\sqrt{20} \sqrt{24}} \\
 \theta &= \cos^{-1} \left[\frac{20}{\sqrt{20} \sqrt{24}} \right] \\
 &= \cos^{-1} \left[\frac{\sqrt{20}}{\sqrt{24}} \right]
 \end{aligned}$$

8. Find the angle between the surfaces $x \log z = y^2 - 1$ and $x^2 y = 2 - z$ at the point $(1, 1, 1)$

Solution: let $\phi_1 = y^2 - x \log z$ and $\phi_2 = x^2 y + z$

$$\begin{aligned}
 \frac{\partial \phi_1}{\partial x} &= -\log z, \quad \frac{\partial \phi_1}{\partial y} = 2y, \quad \frac{\partial \phi_1}{\partial z} = -\frac{x}{z} \\
 \frac{\partial \phi_2}{\partial x} &= 2xy, \quad \frac{\partial \phi_2}{\partial y} = x^2, \quad \frac{\partial \phi_2}{\partial z} = 1 \\
 \nabla \phi_1 &= (-\log z)\vec{i} + 2y\vec{j} - \frac{x}{z}\vec{k} \\
 \nabla \phi_2 &= 2y\vec{j} + x^2\vec{i} + \vec{k}
 \end{aligned}$$

$$\cos \theta = \frac{\nabla \phi_1 \cdot \nabla \phi_2}{|\nabla \phi_1| |\nabla \phi_2|} = \frac{\sqrt{2-1}}{\sqrt{4+1} \sqrt{4+1+1}} = \frac{1}{\sqrt{5}\sqrt{6}}$$

$$\theta = \cos^{-1} \left[\frac{1}{\sqrt{5}\sqrt{6}} \right]$$

9 Find $\nabla^2(r^n)$

$$\begin{aligned} \text{Solution: } \nabla^2(r^n) &= \nabla \cdot \nabla(r^n) = \vec{i} \frac{\partial}{\partial x}(r^n) + \vec{j} \frac{\partial}{\partial y}(r^n) + \vec{k} \frac{\partial}{\partial z}(r^n) \\ &= \vec{i} n r^{n-1} \frac{\partial r}{\partial x} + \vec{j} n r^{n-1} \frac{\partial r}{\partial y} + \vec{k} n r^{n-1} \frac{\partial r}{\partial z} \\ &= n r^{n-2} \left(x \vec{i} + y \vec{j} + z \vec{k} \right) \end{aligned}$$

$$r = \sqrt{x^2 + y^2 + z^2}$$

$$r^2 = x^2 + y^2 + z^2$$

$$2r \frac{\partial r}{\partial x} = 2x \Rightarrow \frac{\partial r}{\partial x} = \frac{x}{r}$$

$$2r \frac{\partial r}{\partial y} = 2y \Rightarrow \frac{\partial r}{\partial y} = \frac{y}{r}$$

$$2r \frac{\partial r}{\partial z} = 2z \Rightarrow \frac{\partial r}{\partial z} = \frac{z}{r}$$

$$\begin{aligned} \therefore \nabla^2(r^n) &= n r^{n-2} \left(x \vec{i} + y \vec{j} + z \vec{k} \right) \\ &= n r^{n-2} \vec{r} \\ &= n r^{n-2} \vec{r} \end{aligned}$$

$$\text{Since } \nabla \cdot (\phi \vec{u}) = \nabla \phi \cdot \vec{u} + \phi \nabla \cdot \vec{u}$$

$$\begin{aligned} \nabla^2(r^n) &= \nabla \cdot (n r^{n-2} \vec{r}) \\ &= n r^{n-2} \nabla \cdot \vec{r} + \nabla(n r^{n-2}) \cdot \vec{r} \end{aligned}$$

$$\begin{aligned} \nabla \cdot \vec{r} &= \frac{\partial}{\partial x}(x) + \frac{\partial}{\partial y}(y) + \frac{\partial}{\partial z}(z) = 1 + 1 + 1 = 3 \end{aligned}$$

$$\begin{aligned} \nabla^2(r^n) &= 3n r^{n-2} + n \nabla(n r^{n-2}) \cdot \vec{r} \\ &= 3n r^{n-2} + n(n-2) r^{n-4} r^2 \\ &= 3n r^{n-2} + n(n-2) r^{n-2} \end{aligned}$$

$$\nabla^2(r^n) = r^{n-2} [n^2 + n] = n(n+1) r^{n-2}$$

10 If $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$ and $r = |\vec{r}|$. Prove that $r^n \vec{r}$ is solenoidal if $n = -3$. Mathematics II

$r^n \vec{r}$ is irrotational for all vectors of n .

Solution: $r^n \vec{r} = r^n (x\vec{i} + y\vec{j} + z\vec{k})$

$$\text{div} (r^n \vec{r}) = \frac{\partial}{\partial x} (r^n x) + \frac{\partial}{\partial y} (r^n y) + \frac{\partial}{\partial z} (r^n z) \dots \dots \dots (1)$$

Now $r^2 = x^2 + y^2 + z^2$

Differentiating partially w.r.to x ,

$$2r \frac{\partial r}{\partial x} = 2x \Rightarrow \frac{\partial r}{\partial x} = \frac{x}{r}$$

Similarly,

$$2r \frac{\partial r}{\partial y} = 2y \Rightarrow \frac{\partial r}{\partial y} = \frac{y}{r}$$

$$2r \frac{\partial r}{\partial z} = 2z \Rightarrow \frac{\partial r}{\partial z} = \frac{z}{r}$$

Now $\frac{\partial}{\partial x} (r^n x) = x \frac{\partial}{\partial r} (r^n) \frac{\partial r}{\partial x} + r^n$

$$= x \cdot n r^{n-1} \frac{x}{r} + r^n$$

$$\frac{\partial}{\partial y} (r^n y) = n r^{n-2} y^2 + r^n$$

$$\frac{\partial}{\partial z} (r^n z) = n r^{n-2} z^2 + r^n$$

From (1) we have

$$\begin{aligned} \text{div} (r^n \vec{r}) &= n r^{n-2} (x^2 + y^2 + z^2) + 3r^n \\ &= n r^n + 3r^n \\ &= (n+3)r^n \end{aligned}$$

The vector $r^n \vec{r}$ is solenoidal if $\text{div} (r^n \vec{r}) = 0$

$$\begin{aligned} \Rightarrow (n+3)r^n &= 0 \\ \Rightarrow n+3 &= 0 \\ \Rightarrow n &= -3 \end{aligned}$$

$\therefore r^n \vec{r}$ is solenoidal only if $n = -3$

Now $\text{curl} (r^n \vec{r}) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ r^n x & r^n y & r^n z \end{vmatrix}$

$$= \sum \vec{i} \left(\frac{\partial}{\partial y} (r^n z) - \frac{\partial}{\partial z} (r^n y) \right)$$

$$= \sum \vec{i} \left(n r^{n-1} \frac{\partial r}{\partial y} z - n r^{n-1} y \frac{\partial r}{\partial z} \right)$$

$$\begin{aligned}
&= \sum_{r=1}^n i \left(nr^{n-1} \frac{y}{r} z - nr^{n-1} y \frac{z}{r} \right) \\
&= \sum_{r=1}^n i \left(nr^{n-2} yz - nr^{n-2} yz \right) \\
&= 0 \\
\text{Curl } (r^n \vec{r}) &= 0 \vec{i} + 0 \vec{j} + 0 \vec{k} = 0
\end{aligned}$$

$$\text{Curl } (r^n \vec{r}) = 0 \text{ for all values of } n$$

Hence $r^n \vec{r}$ is irrotational for all values of n .

11. Prove that $\vec{F} = (y^2 \cos x + z^3) \vec{i} + (2y \sin x - 4) \vec{j} + 3xz^2 \vec{k}$ is irrotational and find its scalar potential

Solution:

$$\begin{aligned}
\text{curl } \vec{F} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y^2 \cos x + z^3 & 2y \sin x - 4 & 3xz^2 \end{vmatrix} \\
&= \vec{i} \left[\frac{\partial}{\partial y} (3xz^2) - \frac{\partial}{\partial z} (2y \sin x - 4) \right] - \vec{j} \left[\frac{\partial}{\partial x} (3xz^2) - \frac{\partial}{\partial z} (y^2 \cos x + z^3) \right] + \vec{k} \left[\frac{\partial}{\partial x} (2y \sin x - 4) - \frac{\partial}{\partial y} (y^2 \cos x + z^3) \right] \\
&= \vec{i} [0 - 0] - \vec{j} [3z^2 - 3z^2] + \vec{k} [2y \cos x - 2y \cos x] = 0
\end{aligned}$$

$\therefore \vec{F}$ is irrotational.

$$\begin{aligned}
\text{To Find } \phi \text{ such that } \vec{F} &= \text{grad } \phi \\
\therefore (y^2 \cos x + z^3) \vec{i} + (2y \sin x - 4) \vec{j} + 3xz^2 \vec{k} &= \vec{i} \frac{\partial \phi}{\partial x} + \vec{j} \frac{\partial \phi}{\partial y} + \vec{k} \frac{\partial \phi}{\partial z}
\end{aligned}$$

Integrating the equation partially w.r.to x, y, z respectively

$$\phi = y^2 \sin x + xz^3 + f_1(y, z)$$

$$\phi = y^2 \sin x - 4y + f_2(x, z)$$

$$\phi = xz^3 + f_3(x, y)$$

$\therefore \phi = y^2 \sin x + xz^3 - 4y + C$, is scalar potential

Q Prove that $\text{div}(\vec{A} \times \vec{B}) = \vec{B} \cdot (\text{curl } \vec{A}) - \vec{A} \cdot (\text{curl } \vec{B})$

Proof : $\text{div}(\vec{A} \times \vec{B}) = \nabla \cdot (\vec{A} \times \vec{B})$

$$\begin{aligned}
&= \sum_{i=1}^3 \vec{i} \cdot \frac{\partial}{\partial x_i} (\vec{A} \times \vec{B}) \\
&= \sum_{i=1}^3 \vec{i} \cdot (\vec{A} \times \vec{B})_{\vec{i}} + \sum_{i=1}^3 \vec{i} \cdot \frac{\partial \vec{A}}{\partial x_i} \times \vec{B} \\
&= - \sum_{i=1}^3 \vec{i} \cdot \vec{B} \times \vec{A}_{\vec{i}} + \sum_{i=1}^3 \vec{i} \cdot \frac{\partial \vec{A}}{\partial x_i} \times \vec{B}
\end{aligned}$$

$$\begin{aligned}
&= - \sum_i \vec{i} \times \frac{\partial \vec{B}}{\partial x_i} \cdot \vec{A} + \sum_i \vec{i} \times \frac{\partial \vec{A}}{\partial x_i} \cdot \vec{B} \\
&= - \text{curl } \vec{B} \cdot \vec{A} + \text{curl } \vec{A} \cdot \vec{B}
\end{aligned}$$

Prove that $\text{curl}(\text{curl } \vec{F}) = \nabla(\nabla \cdot \vec{F}) - \nabla^2 \vec{F}$

Solution:

$$\text{curl}(\text{curl } \vec{F}) = \nabla \times (\nabla \times \vec{F})$$

$$\begin{aligned}
\text{By using } \vec{a} \times (\vec{b} \times \vec{c}) &= \vec{a}(\vec{b} \cdot \vec{c}) - \vec{b}(\vec{a} \cdot \vec{c}) \\
&= \nabla(\nabla \cdot \vec{F}) - (\nabla \cdot \nabla) \vec{F} \\
&= \nabla(\nabla \cdot \vec{F}) - \nabla^2 \vec{F}
\end{aligned}$$

VECTOR INTEGRATION

Line, surface and Volume Integrals

Problems based on line Integral

Example 1: If $\vec{F} = (3x^2 + 6y)\vec{i} - 14yz\vec{j} + 20xz^2\vec{k}$ Evaluate $\int_C \vec{F} \cdot d\vec{r}$ from (0,0,0) to (1,1,1) along the curve $x = t, y = t^2, z = t^3$

Solution: The end points are (0, 0, 0) and (1, 1, 1)

These points correspond to $t = 0$ and $t = 1$

$$\therefore dx = dt, dy = 2t dt, dz = 3t^2 dt$$

$$\begin{aligned}
\int_C \vec{F} \cdot d\vec{r} &= \int_0^1 (3x^2 + 6y)dx - 14yzdy + 20xz^2dz \\
&= \int_0^1 (3t^2 + 6t^2)dt - 14t^5(2tdt) + 20t^7(3t^2dt) \\
&= \int_0^1 (9t^2 - 28t^6 + 60t^9)dt \\
&= \left(3t^3 - 4t^7 + 6t^{10}\right)_0^1 \\
&= [(3 - 4 + 6) - 0] = 5
\end{aligned}$$

Example 2:

Show that $\vec{F} = x^2\vec{i} + y^2\vec{j} + z^2\vec{k}$ is a conservative vector field.

Solution: If $\vec{F}$ is conservative then $\nabla \times \vec{F} = 0$

Mathematics II

$$\text{Now } \nabla \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 & y^2 & z^2 \end{vmatrix} = 0\vec{i} + 0\vec{j} + 0\vec{k} = 0$$

$\therefore \vec{F}$ is a conservative vector field.

Surface Integrals

Definition: Consider a surface S. Let $\vec{n}$ denote the unit outward normal to the surface S. Let R be the projection of the surface x on the XY plane. Let $\vec{f}$ be a vector valued defined in some region containing the surface S. Then the surface integral of $\vec{f}$ is defined to be $\iint_S \vec{f} \cdot \vec{n} ds = \iint_R \vec{f} \cdot \vec{n} \left| \frac{\partial \vec{r}}{\partial x} \times \frac{\partial \vec{r}}{\partial y} \right| dx dy$

Example 1;

Evaluate $\iint_S \vec{f} \cdot \vec{n} ds$ where $\vec{F} = z\vec{i} + x\vec{j} - y^2z\vec{k}$ and S is the surface of the cylinder $x^2 + y^2 = 1$ included in the first octant between the planes $z = 0$ and $z = 2$.

Solution: Given $\vec{F} = z\vec{i} + x\vec{j} - y^2z\vec{k}$

$$\phi = x^2 + y^2 - 1$$

$$\begin{aligned} \nabla \phi &= 2x\vec{i} + 2y\vec{j} \\ |\nabla \phi| &= \sqrt{4x^2 + 4y^2} \\ &= 2\sqrt{x^2 + y^2} \\ &= 2 \end{aligned}$$

$$\begin{aligned} \text{The unit normal } \hat{n} \text{ to the surface} &= \frac{\nabla \phi}{|\nabla \phi|} \\ &= \frac{2x\vec{i} + 2y\vec{j}}{2} = x\vec{i} + y\vec{j} \end{aligned}$$

$$\vec{F} \cdot \hat{n} = (z\vec{i} + x\vec{j} - y^2z\vec{k}) \cdot (x\vec{i} + y\vec{j}) = xz + xy$$

$$\text{Now } \iint_S \vec{F} \cdot \hat{n} dS = \iint_R \vec{F} \cdot \hat{n} \frac{dy dz}{|\hat{n} \cdot \vec{i}|}$$

Where R is the projection of S on YZ plane.

$$\begin{aligned}
&= \iint_R (xz + xy) \frac{dy \, dz}{x} \quad [\because \hat{n} \cdot \hat{i} = (x\hat{i} + y\hat{j}) \cdot \hat{i} = x] \\
&= \iint_R (z + y) \, dy \, dz \\
&= \int_0^2 \int_0^1 (z + y) \, dy \, dz \\
&= \int_0^2 \left[zy + \frac{y^2}{2} \right]_0^1 dz = \int_0^2 \left(z + \frac{1}{2} \right) dz \\
&= \left[\frac{z^2}{2} + \frac{1}{2}z \right]_0^2 \\
&= \left(\frac{4}{2} + \frac{2}{2} \right) - (0 + 0) \\
&= (2 + 1) = 3.
\end{aligned}$$

Example Evaluate $\iint_S \vec{F} \cdot \hat{n} \, dS$ where $\vec{F} = 18z\vec{i} - 12\vec{j} + 3y\vec{k}$ as S is the part of the plane $2x + 3y + 6z = 12$ which is in the first octant.

Solution : Given : $\vec{F} = 18z\vec{i} - 12\vec{j} + 3y\vec{k}$

$$\text{Let } \phi = 2x + 3y + 6z - 12$$

$$\hat{n} = \frac{\nabla \phi}{|\nabla \phi|}$$

$$\begin{aligned}
\nabla \phi &= \vec{i} \frac{\partial \phi}{\partial x} + \vec{j} \frac{\partial \phi}{\partial y} + \vec{k} \frac{\partial \phi}{\partial z} \\
&= 2\vec{i} + 3\vec{j} + 6\vec{k}
\end{aligned}$$

$$|\nabla \phi| = \sqrt{4 + 9 + 36} = \sqrt{49} = 7$$

$$\hat{n} = \frac{\nabla \phi}{|\nabla \phi|} = \frac{2\vec{i} + 3\vec{j} + 6\vec{k}}{7}$$

$$\vec{F} \cdot \hat{n} = (18z\vec{i} - 12\vec{j} + 3y\vec{k}) \cdot \left(\frac{2\vec{i} + 3\vec{j} + 6\vec{k}}{7} \right)$$

$$= \frac{36z - 36 + 18y}{7}$$

$$= \frac{1}{7} \left[36 \left(\frac{12 - 2x - 3y}{6} \right) - 36 + 18y \right] \quad \left[\because z = \frac{12 - 2x - 3y}{6} \right]$$

$$= \frac{1}{7} [6(12 - 2x - 3y) - 36 + 18y] = \frac{1}{7} [36 - 12x]$$

$$\iint_S \vec{F} \cdot \hat{n} dS = \iint_R \frac{36 - 12x}{7} \frac{dx dy}{|\hat{n} \cdot \vec{k}|}$$

$$\hat{n} \cdot \vec{k} = \frac{2\vec{i} + 3\vec{j} + 6\vec{k}}{7} \cdot \vec{k} = \frac{6}{7}$$

$$\therefore \iint_S \vec{F} \cdot \hat{n} dS = \iint_R \frac{36 - 12x}{7} \frac{7}{6} dx dy$$

$$= \iint_R (6 - 2x) dx dy$$

Given plane equation is $2x + 3y + 6z = 12$ in XOY area $z = 0$.

$$2x + 3y = 12$$

$$3y = 12 - 2x$$

$$y = \frac{12 - 2x}{3}$$

$\therefore y$ varies from 0 to $\frac{12 - 2x}{3}$ in X plane $y = 0, z = 0$

$$\therefore 2x = 12$$

$$x = 6$$

$\therefore x$ varies from 0 to 6

$$= \int_0^6 \int_0^{\frac{12 - 2x}{3}} (6 - 2x) dy dx$$

$$= \int_0^6 (6y - 2xy)_0^{\frac{12 - 2x}{3}} dx$$

$$= \int_0^6 \left[6 \left(\frac{12 - 2x}{3} \right) - 2x \left(\frac{12 - 2x}{3} \right) \right] dx$$

$$\begin{aligned}
&= \int_0^6 \left(24 - 4x - 8x + \frac{4x^2}{3} \right) dx \\
&= \int_0^6 \left(24 - 12x + \frac{4x^2}{3} \right) dx \\
&= \left[24x - 6x^2 + \frac{4x^3}{9} \right]_0^6 = 24.
\end{aligned}$$

VOLUME INTEGRALS

The volume integral of $F(x, y, z)$ over a region enclosing a volume V is given by $\iiint_V F(x, y, z) dV$ or $\iiint_V F(x, y, z) dx dy dz$.

PROBLEMS BASED ON VOLUME INTEGRALS

Example 2.3.16. If $\vec{F} = (2x^2 - 3z)\vec{i} - 2xy\vec{j} - 4x\vec{k}$, evaluate $\iiint_V \nabla \times \vec{F} dV$ where

V is the region bounded by $x = 0$, $y = 0$, $z = 0$ and $2x + 2y + z = 4$.

$$\begin{aligned}
\text{Solution : } \nabla \times \vec{F} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2x^2 - 3z & -2xy & -4x \end{vmatrix} \\
&= \vec{i}(0) + \vec{j}(-3 + 4) + \vec{k}(-2y - 0) \\
&= \vec{j} - 2y\vec{k}
\end{aligned}$$

$$\begin{aligned}
\therefore \iiint_V \nabla \times \vec{F} dV &= \int_0^2 \int_0^{2-x} \int_0^{4-2x-2y} (\vec{j} - 2y\vec{k}) dz dy dx \\
&= \int_0^2 \int_0^{2-x} (z\vec{j} - 2yz\vec{k})_0^{4-2x-2y} dy dx \\
&= \int_0^2 \int_0^{2-x} [4 - 2x - 2y]\vec{j} - 2y(4 - 2x - 2y)\vec{k} dy dx \\
&= \int_0^2 \left[(4y - 2xy - y^2)\vec{j} - \left(4y^2 - 2xy^2 - \frac{4y^3}{3} \right) \vec{k} \right]_0^{2-x} dx
\end{aligned}$$

$$\begin{aligned}
&= \int_0^2 \left[\left[4(2-x) - 2x(2-x) - (2-x)^2 \right] j \right. \\
&\quad \left. - \left[4(2-x)^2 - 2x(2-x)^2 - \frac{4}{3}(2-x)^3 \right] k \right] dx \\
&= \int_0^2 (8 - 4x - 4x + 2x^2 - 4 - x^2 + 4x) j^{\rightarrow} \\
&\quad - [(16 - 16x + 4x^2 - 8x + 8x^2 - 2x^3) \\
&\quad - \frac{4}{3}(8 - 12x + 6x^2 - x^3) k] dx \\
&= \int_0^2 \left[(4 - 4x + x^2) j^{\rightarrow} - \frac{k^{\rightarrow}}{3} (16 - 24x + 12x^2 - 2x^3) \right] dx \\
&= \left[4x - 2x^2 + \frac{x^3}{3} \right] j^{\rightarrow} - \frac{k^{\rightarrow}}{3} \left[16x - 12x^2 + 4x^3 - \frac{x^4}{2} \right]_0^2 \\
&= \left(8 - 8 + \frac{8}{3} \right) j^{\rightarrow} - \frac{k^{\rightarrow}}{3} (32 - 48 + 32 - 8) = \frac{8}{3} (j^{\rightarrow} - k^{\rightarrow})
\end{aligned}$$

INTEGRAL THEOREMS

- (i) Gauss's divergence theorem
- (ii) Stoke's theorem
- (iii) Green's theorem in the plane

Green's Theorem

Statement:

If $M(x,y)$ and $N(x,y)$ are continuous functions with continuous partial derivatives in a region R of the xy plane bounded by a simple closed curve C , then

$$\oint_C Mdx + Ndy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) \cdot dxdy, \text{ where } C \text{ is the curve described in the}$$

positive direction.

Verify Green's theorem in a plane for the integral $\int_c (x - 2y)dx + xdy$

taken around the circle $x^2 + y^2 = 4$

Solution: Green's theorem gives

$$\int_c Mdx + Ndy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dxdy$$

Consider $\int_c (x - 2y)dx + xdy$

$$M = x - 2y \quad N = x$$

$$\frac{\partial M}{\partial y} = -2, \quad \frac{\partial N}{\partial x} = 1$$

$$\therefore \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dxdy$$

$$\iint_R (1 + 2) dxdy = 3 \iint_R dxdy$$

$$= 3[\text{Area of the circle}]$$

$$= 3\pi r^2$$

$$= 3 \cdot \pi \cdot 4$$

$$= 12\pi \quad \dots\dots\dots(1)$$

Now $\int Mdx + Ndy$

We know that the parametric equation of the circle $x^2 + y^2 = 4$

$$x = 2 \cos \theta \quad y = 2 \sin \theta$$

$$dx = -2 \sin \theta d\theta, \quad dy = 2 \cos \theta d\theta$$

$$\therefore Mdx + Ndy = (x - 2y)dx + xdy$$

$$= (2 \cos \theta - 4 \sin \theta)(-2 \sin \theta d\theta) + 2 \cos \theta (2 \cos \theta) d\theta$$

$$= -2 \cos \theta \sin \theta + 8 \sin^2 \theta + 4 \cos^2 \theta d\theta$$

Where θ varies from 0 to 2π

$$\therefore \int_c Mdx + Ndy = \int_0^{2\pi} (-2 \cos \theta \sin \theta + 4 \sin^2 \theta + 4 \cos^2 \theta) d\theta$$

$$= \int_0^{2\pi} (-\sin 2\theta + 4 \frac{1 - \cos 2\theta}{2} + 4) d\theta$$

$$= \int_0^{2\pi} (-\sin 2\theta + 6 - 2 \cos 2\theta) d\theta$$

$$= \left[\frac{\cos 2\theta}{2} + 6\theta - \frac{2 \sin 2\theta}{2} \right]_0^{2\pi}$$

$$= \frac{1}{2} + 12\pi - \frac{1}{2} = 12\pi \quad \dots\dots\dots(2)$$

$\therefore$ From (1) and (2)

$$\int_c Mdx + Ndy = \iint_R \left[\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right] dxdy$$

Hence Green's Theorem is verified.

Example 2

Using Green's theorems find the area of a circle of radius r .

Solution: By Green's theorem we know that

$$\text{Area enclosed by } C = \frac{1}{2} \int_C xdy - ydx$$

The parametric equation of a circle of radius r is $x = r \cos \theta$, $y = r \sin \theta$

Where $0 \leq \theta \leq 2\pi$

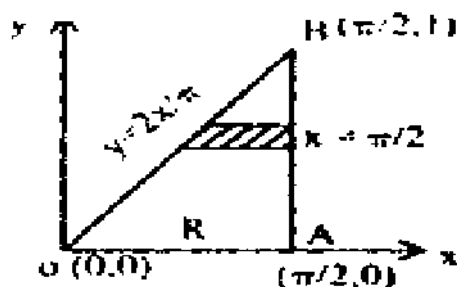
$$\begin{aligned} \therefore \text{Area of the circle} &= \frac{1}{2} \int_0^{2\pi} r \cos \theta (r \cos \theta) - r \sin \theta (-r \sin \theta) d\theta \\ &= \frac{1}{2} \int_0^{2\pi} (r^2 \cos^2 \theta + r^2 \sin^2 \theta) d\theta \\ &= \frac{1}{2} \int_0^{2\pi} r^2 d\theta \\ &= \frac{1}{2} r^2 [\theta]_0^{2\pi} = \pi r^2 \end{aligned}$$

Example 3:

Evaluate $\int_c [(\sin x - y)dx - \cos x dy]$ where c is the triangle with

vertices $(0,0)$, $(\frac{\pi}{2}, 0)$ and $(\frac{\pi}{2}, 1)$

$$\begin{aligned} \text{Solution: Equation of OB is } \frac{y-0}{1-0} &= \frac{x-0}{\frac{\pi}{2}-0} \\ \Rightarrow y &= \frac{2x}{\pi} \end{aligned}$$



By Green's theorem $\int_C Mdx + Ndy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy$

Here $M = \sin x - y, \frac{\partial M}{\partial y} = -1$

$N = -\cos x, \frac{\partial N}{\partial x} = \sin x$

$$\therefore \int_C [(\sin x - y)dx - \cos x dy] = \iint_R (\sin x + 1) dx dy$$

In the region R, x varies from $x = \frac{\pi y}{2}$ to $\frac{\pi}{2}$ and y varies from $y = 0$ to $y = 1$

$$\begin{aligned} \therefore \int_C (\sin x - y)dx - \cos x dy &= \int_0^1 \int_{\frac{\pi y}{2}}^{\frac{\pi}{2}} (\sin x + 1) dx dy \\ &= \int_0^1 \left[-\cos x + x \right]_{\frac{\pi y}{2}}^{\frac{\pi}{2}} dy \\ &= \int_0^1 \left[\cos \frac{\pi y}{2} + \frac{\pi y}{2} - \frac{\pi y^2}{2} \right] dy \\ &= \left[\sin \frac{\pi y}{2} + \frac{\pi y^2}{2} - \frac{\pi y^3}{6} \right]_0^1 \\ &= \frac{2}{\pi} + \frac{\pi}{2} - \frac{\pi}{4} = \frac{2}{\pi} + \frac{\pi}{2} \end{aligned}$$

Example 4

Verify Green's theorem in the plane for

$\int_C (3x^2 - 8y^2)dx + (4y - 6xy)dy$ where C is the boundary of the region defined

by

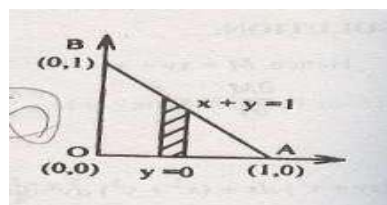
$x = 0, y = 0, x + y = 1$

Solution: We have to prove that

$$\int_c Mdx + Ndy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy$$

$$M = 3x^2 - 8y^2, N = 4y - 6xy$$

$$\frac{\partial M}{\partial y} = -16y, \frac{\partial N}{\partial x} = -6y$$



By Green's theorem in the plane

$$\int_c Mdx + Ndy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy$$

$$= \int_0^1 \int_0^{1-x} (10y) dy dx$$

$$= 10 \int_0^1 \left[\frac{y^2}{2} \right]_0^{1-x} dx$$

$$= 5 \int_0^1 (1-x)^2 dx$$

$$= 5 \left[\frac{(1-x)^3}{-3} \right]_0^1 = \frac{5}{3}$$

Consider $\int Mdx + Ndy = \int_{OA} + \int_{AB} + \int_{BO}$

Along OA, $y=0$, x varies from 0 to 1

$$\therefore \int_{OA} Mdx + Ndy = \int_0^1 3x^2 dx = \left[x^3 \right]_0^1 = 1$$

Along AB, $y = 1 - x \Rightarrow dy = -dx$ and x varies from 1 to 0.

$$\therefore \int_{AB} Mdx + Ndy = \int_1^0 \left[3x^2 - 8(1-x)^2 - 4(1-x) + 6x(1-x) \right] dx$$

$$= \left[\frac{3x^3}{3} - \frac{8(1-x)^3}{-3} - \frac{4(1-x)^2}{-2} + 3x^2 - 2x^3 \right]_1^0$$

$$= \frac{8}{3} + 2 - 1 - 3 + 2 = \frac{8}{3}$$

STOKE'S THEOREM

If S is an open surface bounded by a simple closed curve C and if a vector function $\vec{F}$ is continuous and has continuous partial derivatives in S and on C , then $\oint_C \vec{F} \cdot d\vec{r} = \iint_S \text{curl } \vec{F} \cdot \vec{n} \, ds$ where $\vec{n}$ is the unit vector normal to the surface (ie) The surface integral of the normal component of $\text{curl } \vec{F}$ is equal to the integral of the tangential component of $\vec{F}$ taken around C .

Example 1

Verify Stoke's theorem for $\vec{F} = (2x - y)\vec{i} - yz^2\vec{j} - y^2z\vec{k}$ where S is the upper half of the sphere $x^2 + y^2 + z^2 = 1$ and C is the circular boundary on $z = 0$ plane.

Solution: By Stoke's theorem $\oint_C \vec{F} \cdot d\vec{r} = \iint_S \text{curl } \vec{F} \cdot \vec{n} \, ds$

$$\vec{F} = (2x - y)\vec{i} - yz^2\vec{j} - y^2z\vec{k}$$

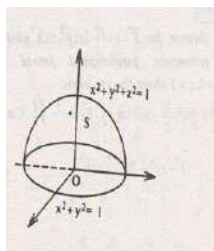
$$\text{curl } \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2x - y & -yz^2 & -y^2z \end{vmatrix}$$

$$= \vec{i}[-2yz + 2yz] - \vec{j}(0 - 0) + \vec{k}(0 + 1) = \vec{k}$$

Here $\vec{n} = \vec{k}$ since C is the circular boundary on $z = 0$ plane

$$\therefore \iint_S \text{curl } \vec{F} \cdot \vec{n} \, ds = \iint_S dxdy$$

$$= \pi (1)^2 = \pi \dots\dots\dots(1)$$



ON $z = 0$, $\oint_C \vec{F} \cdot d\vec{r} = \iint_S \text{curl } \vec{F} \cdot \vec{n} \, ds$

On C , $x = \cos\theta$, $y = \sin\theta$

$dx = -\sin\theta d\theta$, $dy = \cos\theta d\theta$

θ varies from 0 to 2π

$$\begin{aligned}
\therefore \int_c \vec{F} \cdot d\vec{r} &= \int_0^{2\pi} (2 \cos \theta - \sin \theta) (-\sin \theta) d\theta \\
&= - \int_0^{2\pi} (2 \cos \theta \sin \theta) d\theta + \int_0^{2\pi} \sin^2 \theta d\theta \\
&= - \int_0^{2\pi} \sin 2\theta d\theta + \int_0^{2\pi} \frac{1 - \cos 2\theta}{2} d\theta \\
&= - \left[-\frac{\cos 2\theta}{2} \right]_0^{2\pi} + \left[\frac{\theta}{2} - \frac{\sin 2\theta}{2} \right]_0^{2\pi} \\
&= - \left[-\frac{1}{2} + \frac{1}{2} \right] + \left[\pi - 0 \right] = \pi \dots\dots\dots(2)
\end{aligned}$$

$\therefore$ From (1) and (2)

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_S \text{curl } \vec{F} \cdot \vec{n} \, ds$$

Hence stoke's theorem is verified

Example 2

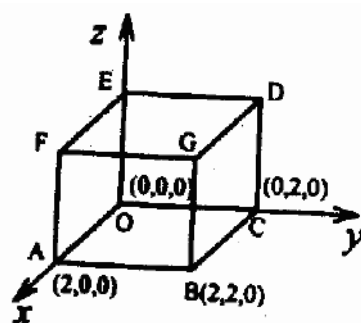
Verify stoke's theorem for $\vec{F} = (y - z + 2)\vec{i} + (yz + 4)\vec{j} - xz\vec{k}$ where s is the surface of the cube $x = 0, x = 2, y = 0, y = 2, z = 0$ and $z = 2$ above the xy - plane.

Solution:

By Stoke's theorem

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_S \text{curl } \vec{F} \cdot \vec{n} \, ds$$

$$\begin{aligned}
\text{Given } \vec{F} &= (y - z + 2)\vec{i} + (yz + 4)\vec{j} - xz\vec{k} \quad \text{curl } \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y - z + 2 & yz + 4 & -xz \end{vmatrix} \\
&= \vec{i}[0 - y] - \vec{j}[z + 1] + \vec{k}[0 - 1] \\
&= -y\vec{i} + \vec{j}[z - 1] - \vec{k}
\end{aligned}$$



Surface	Equation	$\vec{n}$	ds	$\text{Curl } \vec{F} \cdot \vec{n}$
$ABEF (s_1)$	$x = 2$	$\vec{i}$	$dy dz$	$-y$
$OCDG (s_2)$	$x = 0$	$-\vec{i}$	$dy dz$	y
$BCDE (s_3)$	$y = 2$	$\vec{j}$	$dx dz$	$-(z-1)$
$OAFG (s_4)$	$y = 0$	$-\vec{j}$	$dx dz$	$-(z-1)$
$DEFG (s_5)$	$z = 2$	$\vec{k}$	$dx dy$	-1

$$\therefore \iiint_S \text{curl } \vec{F} \cdot \vec{n} dx dy dz = \iiint_{s_1} + \iiint_{s_2} + \iiint_{s_3} + \iiint_{s_4} + \iiint_{s_5}$$

From curl $\vec{F} \cdot \vec{n}$ values all the first four integrals vanishes.

$$\therefore \iiint_{s_5} \text{curl } \vec{F} \cdot \vec{n} dx dy dz = - \int_0^2 \int_0^2 dx dy = -4$$

$$\therefore \int \int_S \text{curl } \vec{F} \cdot \vec{n} ds = -4 \quad \dots (1)$$

The boundary curve c is the face $OABC$, which is a square with sides, OA, AB, BC and CO .

$$\therefore \int_c \vec{F} \cdot d\vec{r} = \int_{OA} \vec{F} \cdot d\vec{r} + \int_{AB} \vec{F} \cdot d\vec{r} + \int_{BC} \vec{F} \cdot d\vec{r} + \int_{CO} \vec{F} \cdot d\vec{r}$$

$$\vec{F} \cdot d\vec{r} = (y - z + 2) dx + (yz + 4) dy - xz dz$$

$$= (y + 2) dx + 4dy \quad [\text{since } z = 0]$$

Along OA , $y = 0$, $dy = 0$, x varies from 0 to 2.

$$\therefore \int_{OA} \vec{F} \cdot d\vec{r} = \int_0^2 2dx = 4$$

Along AB , $x = 2$, $dx = 0$, y varies from 0 to 2

$$\int_{AB} \vec{F} \cdot d\vec{r} = \int_0^2 4 dy = 8$$

Along BC , $y = 2$, $dy = 0$, x varies from 2 to 0

$$\int_{BC} \vec{F} \cdot d\vec{r} = \int_2^0 4 dx = -8$$

Along CO , $x = 0$, $dx = 0$, y varies from 2 to 0

$$\int_{CO} \vec{F} \cdot d\vec{r} = \int_2^0 4dy = -8$$

$$\therefore \int_c \vec{F} \cdot d\vec{r} = 4 + 8 - 8 - 8 = -4 \quad \dots (2)$$

From (1) and (2)

$$\text{Hence} \quad \int_c \vec{F} \cdot d\vec{r} = \int \int_S \text{curl } \vec{F} \cdot \vec{n} ds$$

Hence Stoke's theorem is verified.

Example 3:

Verify Stoke's theorem for $\vec{F} = y\vec{i} + z\vec{j} + x\vec{k}$ where S is the upper half surface of the sphere $x^2 + y^2 + z^2 = 1$ and C is its boundary.

Solution: By stoke's theorem

$$\iint_S \text{curl } \vec{F} \cdot \vec{n} ds = \int_C \vec{F} \cdot d\vec{r}$$

$$\vec{F} = y\vec{i} + z\vec{j} + x\vec{k}$$

$$\begin{aligned} \text{Curl } \vec{F} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y & z & x \end{vmatrix} \\ &= \vec{i}[0-1] - \vec{j}[1-0] + \vec{k}[0-1] = -\vec{i} - \vec{j} - \vec{k} \end{aligned}$$

$$\text{Now } \phi = x^2 + y^2 + z^2 - 1$$

$$\nabla\phi = 2x\vec{i} + 2y\vec{j} + 2z\vec{k}$$

$$|\nabla\phi| = \sqrt{4x^2 + 4y^2 + 4z^2} = 2$$

$$\therefore \vec{n} = \frac{\nabla\phi}{|\nabla\phi|} = x\vec{i} + y\vec{j} + z\vec{k}$$

$$\therefore \text{curl } \vec{F} \cdot \vec{n} = -x - y - z$$

S be the projection on XOY plane gives

$$ds = \frac{dx dy}{|\vec{n} \cdot \vec{k}|} = \frac{dx dy}{z}$$

$$\begin{aligned} \therefore \iint_S (-x - y - z) \frac{dx dy}{z} &= \iint_S \frac{(-x - y)}{z} dx dy - \iint_S dx dy \\ &= \iint_S \frac{(-x - y)}{\sqrt{1 - x^2 - y^2}} dx dy - \iint_S dx dy \\ &= I_1 - I_2 \end{aligned}$$

$$\text{Now } I_1 = \iint_S \frac{-x - y}{\sqrt{1 - x^2 - y^2}} dx dy$$

By changing to polar coordinates

$$x = r \cos \theta, y = r \sin \theta \quad dx dy = r dr d\theta$$

r varies from 0 to 1

θ varies from 0 to 2π

$$\begin{aligned}
 I_1 &= \int_0^{2\pi} \int_0^1 \frac{\cos \theta + \sin \theta}{\sqrt{1-r^2}} r^2 dr d\theta \\
 &= - \int_0^1 \frac{r^2}{\sqrt{1-r^2}} dr \int_0^{2\pi} (\cos \theta + \sin \theta) d\theta \\
 &= - \int_0^1 \frac{r^2}{\sqrt{1-r^2}} dr [\sin \theta - \cos \theta]_0^{2\pi} = - \int_0^1 \frac{r^2}{\sqrt{1-r^2}} dr [-1 + 1] = 0
 \end{aligned}$$

$$\text{Now } I_2 = \iint_S dx dy$$

$$= \text{Area of the closed circle } x^2 + y^2 = 1 = \pi$$

$$\therefore I_1 - I_2 = 0 - \pi = -\pi$$

$$\therefore \iint_S \text{curl } \vec{F} \cdot \vec{n} ds = -\pi \quad \dots(1)$$

$$\vec{F} \cdot d\vec{r} = ydx + zdy + xdz$$

The equation of the boundary circle is $x^2 + y^2 = 1$

since $z=0$, $dz=0$, we have $\vec{F} \cdot d\vec{r} = ydx$

put $x = \cos \theta$, $y = \sin \theta$

$$dx = -\sin \theta d\theta, \quad dy = \cos \theta d\theta$$

θ varies from 0 to 2π

$$\begin{aligned}
 \int_C \vec{F} \cdot d\vec{r} &= - \int_0^{2\pi} \sin \theta^2 d\theta = - \frac{1}{2} \int_0^{2\pi} (1 - \cos 2\theta) d\theta \\
 &= - \frac{1}{2} \left[\theta - \frac{\sin 2\theta}{2} \right]_0^{2\pi} \\
 &= -\pi
 \end{aligned}$$

... (2)

$\therefore$ From (1) and (2)

$$\int_{\gamma} \vec{F} \cdot d\vec{r} = \iiint_{\gamma} \text{curl } \vec{F} \cdot \vec{n} \, ds$$

Gauss Divergence theorem

Statement:

The surface integral of the normal component of a vector function F over a closed surface S enclosing volume V is equal to the volume integral of the divergence of F taken throughout the volume V ,

$$\iiint_S \vec{F} \cdot \vec{n} \, ds = \iiint_V \nabla \cdot \vec{F} \, dv$$

Evaluate $\iiint x^3 dy dz + x^2 y dz dx + x^2 z dx dy$ over the surface bounded by $z = 0, z = h, x^2 + y^2 = a^2$

Solution:

$$\iiint_S (F_1 dy dz + F_2 dz dx + F_3 dx dy) = \iiint_V \left(\frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z} \right) dx dy dz$$

$$\text{Here } F_1 = x^3, F_2 = x^2 y, F_3 = x^2 z$$

$$\therefore \frac{\partial F_1}{\partial x} = 3x^2, \quad \frac{\partial F_2}{\partial y} = x^2, \quad \frac{\partial F_3}{\partial z} = x^2$$

$$\therefore \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z} = 5x^2$$

$$\therefore \iiint_S F_1 dy dz + F_2 dz dx + F_3 dx dy = \iiint_V 5x^2 dx dy dz$$

$$= \int_{z=0}^h \int_{y=-a}^a \int_{x=\sqrt{a^2-y^2}}^{\sqrt{a^2-y^2}} 5x^2 dx dy dz$$

$$= \int_0^h \int_{-a}^a \left[\frac{10x^3}{3} \right]_0^{\sqrt{a^2-y^2}} dy dz$$

$$= \int_0^h \int_{-a}^a \frac{10}{3} (a^2 - y^2)^{3/2} dy dz$$

$$= \frac{20}{3} \int_0^h \int_0^a (a^2 - y^2)^{3/2} dy dz$$

$$\text{Put } x = a \cos \theta, y = a \sin \theta$$

$$dy = a \cos \theta d\theta$$

$$\Rightarrow \theta \text{ varies from } 0 \text{ to } \pi/2$$

$$\therefore \iiint_S F_1 dy dz + F_2 dz dx + F_3 dx dy$$

$$= \frac{20}{3} \int_0^h \int_0^{\pi/2} (a^2 - a^2 \sin^2 \theta)^{3/2} a \cos \theta d\theta dz$$

$$= \frac{20a^4}{3} \int_0^h \int_0^{\pi/2} \cos^4 \theta d\theta dz$$

$$= \frac{20a^4}{3} \int_0^h \frac{3\pi}{16} dz = \frac{5}{4} a^4 \pi \int_0^h dz$$

$$= \frac{5a^4\pi}{4} h$$

$$\int_0^{\frac{\pi}{4}} \cos^4 \theta d\theta = \frac{3}{4} \frac{1}{2} \frac{\pi}{2} = \frac{3\pi}{16}$$

Evaluate $\iint_s \vec{F} \cdot \vec{n} ds$ where $\vec{F} = 4xz \vec{i} - y^2 \vec{j} + yz \vec{k}$ and s is the surface of

the cube bounded by $x=0, x=a, y=0, y=a, z=0, z=a$ [$x=0, x=1, y=0, y=1, z=0, z=1$ means $a=1$]

SOLUTION: Given $\vec{F} = 4xz \vec{i} - y^2 \vec{j} + yz \vec{k}$

$$\text{div } \vec{F} = 4z - 2y + y = 4z - y$$

x ; varies from 0 to a

y ; varies from 0 to a

z ; varies from 0 to a

$$\iint_s \vec{F} \cdot \vec{n} ds = \iiint_v \text{div } \vec{F} dv$$

$$= \int_0^a \int_0^a \int_0^a (4z - y) dx dy dz$$

$$= \int_0^a \int_0^a [4zx - yx]_0^a dy dz$$

$$= \int_0^a \int_0^a [4za - ya] dy dz$$

$$= \int_0^a \left[4azy - \frac{ay^2}{2} \right]_0^a dz = \int_0^a \left(4a^2z - \frac{a^3}{2} \right) dz$$

$$= \left[2a^2z^2 - \frac{a^3z}{2} \right]_0^a = 2a^4 - \frac{a^4}{2}$$